Question Bank C Programming Level - 1

1. Craft a program that prompts the user to input a student's marks(6 marks). The program should then evaluate and print the category of the student's performance based on grade as (Grade A to F).
   Percentage >= 95% A
   Percentage >= 85% B
   Percentage >= 75% C
   Percentage >= 65% D
   Percentage >= 45% E
   Percentage < 45% F

2. Craft a program where users input marks for five subjects Physics, Chemistry, Biology, Mathematics, and C

3. Craft a program where users input marks for five subjects—Physics, Chemistry, Biology, Mathematics, and Computer. The program swiftly calculates the average and assigns a grade: A, B, C, D, E, or F. Also prints the average value in two decimal points.
   Percentage >= 90% A
   Percentage >= 80% B
   Percentage >= 70% C
   Percentage >= 60% D
   Percentage >= 50% E
   Percentage < 50% F

4. For a calendar application, users input a month number(1-12). The program calculates and prints the number of days in the specified month and also prints the quadrant that month occurs.

5. In a date-related application, users input a number (1-7). The program efficiently determines and prints the corresponding day of the week. This allows users to instantly identify the day associated with a given week number within the application. Also, verify whether the day is a weekday or a holiday.

6. Craft a program that prompts the user to input the number of steps they took in a day and time(in minutes) taken for the activity. The program should then categorize their fitness level.

7. Your task is to develop a program that prompts the user to input the number of hours worked by an employee in a week and their hourly wage rate. The program should then calculate the total salary of the employee according to the following conditions:
   For first 8 hours , if male wage will be Rs.80 per hour
   If female wage will be Rs.60 per hour.
   If more than 8 hours, each extra hour(after the first 8 hour) will be paid with Rs.100 per hour.

8. Craft a program that prompts the user to input two integers. The program should then check whether both the inputs are equal or not using bitwise operators.

9. You are tasked with creating a simple menu-driven calculator. The program should display a menu with options for basic arithmetic operations: addition, subtraction, multiplication, and division.

10. Your task is to create a program that prompts users to input their age. Based on their age, the program will suggest suitable food options tailored to their age group. The program should adhere to the following guidelines:
    1. Prompt the user to input their age.
    2. Use if-else statements to categorise users into specific age groups.
    3. Provide food recommendations based on each age group. For example:
       - For children aged 5 and below, suggest mashed vegetables or soft fruits.
       - For teenagers aged 13 to 18, suggest burgers or pasta.
       - For adults aged 51 and above, suggest soup or grilled fish.
    4. Ensure that the program handles all possible age ranges and provides appropriate food suggestions for each category.

5. Display the recommended food options to the user based on their age group.
6. Implement error handling to address any invalid inputs, such as negative ages or non-numeric entries.

11. The goal of the program is to take user input for a month number and, based on the input, display the name of the month and the number of days it has. If the user inputs the month as February (month number 2), the program should also ask for the year, perform a leap year calculation, and display the appropriate number of days, and print .

12. Craft a program that prompts the user to enter a number. The program should analyse the number and display whether it is positive, negative, or zero.

13. Your task is to create a program that prompts users to input their age. Based on their age, the program will suggest suitable food options tailored to their age group. The program should adhere to the following guidelines:
    1. Prompt the user to input their age.
    2. Use if-else statements to categorise users into specific age groups.
    3. Provide food recommendations based on each age group. For example:
        - For children aged 5 and below, suggest mashed vegetables or soft fruits.
        - For teenagers aged 13 to 18, suggest burgers or pasta.
        - For adults aged 51 and above, suggest soup or grilled fish.
    4. Ensure that the program handles all possible age ranges and provides appropriate food suggestions for each category.
    5. Display the recommended food options to the user based on their age group.
    6. Implement error handling to address any invalid inputs, such as negative ages or non-numeric entries.

14. In the world of entertainment, ticket prices vary based on age. Your task is to create a program that calculates the ticket price for a movie based on the age of the viewer as Children, Teenagers, Adult, Seniors (age 61 and above)

15. Create a program to calculate if Ram's chocolates will last for the duration he plans to eat them, based on input values of total chocolates, consumption per day, and planned days. Output 1 if they last, 0 if they run out."

16. Craft a program that prompts the user to input the quantities of pen, notebook and pencils they wish to purchase. The cost per pen is 20 units, the cost per notebook is 70 units and the cost per pencil is 9 units .The program should then calculate and display the total amount for the shopper.

17. Your task is to create a program that enables customers to select their preferred chocolate flavour from a menu using numeric choices. The program should follow these guidelines:
    1. Display a menu of available chocolate flavours, along with corresponding numeric choices (e.g., 1. Milk Chocolate, 2. Dark Chocolate, 3. White Chocolate).
    2. Prompt the user to input their choice by entering the corresponding number.
    3. Use a switch statement to process the user's choice and display a message based on the selected flavour:
        - For Milk Chocolate, display a message about its creamy taste.
        - For Dark Chocolate, display a message about its rich flavour.
        - For White Chocolate, display a message about its sweet and smooth texture.
        - If the user enters an invalid choice (a number other than 1, 2, or 3), display an error message.
    4. Ensure the program provides a user-friendly experience and handles invalid inputs gracefully.
    5. Upon receiving the user's selection, display a message confirming their choice and encouraging them to enjoy their selected chocolate flavor.

18. Your task is to create a C program that prompts users to input the number of symptoms experienced by a patient. The program will then assess the severity of the patient's condition and assign a priority level based on the following criteria:
    1. Patients experiencing 5 or more symptoms are considered to be in critical condition and require immediate attention with high priority.
    2. Patients experiencing 3 or 4 symptoms are considered to be in moderate condition and require attention soon with medium priority.
    3. Patients experiencing fewer than 3 symptoms are considered to be in stable condition and can wait for a while with low priority.
    4. Implement error handling to ensure that the user enters a valid number of symptoms.
    5. Display the assigned priority level to the user based on the severity of the patient's symptoms.

19. The program efficiently calculates the Gross salary, incorporating specific allowances based on conditions:
    If Basic Salary is <= 70000: HRA = 30%, DA = 80%
    This ensures immediate and accurate feedback on the employee's Gross salary within the application.

20. An instructor tries to make a connection between the direction and the angles written on a circle. Her initial goal is to teach them the relationship between the angles that represent East, North, West, and South, respectively. To visually represent the direction that corresponds to the angle entered, write a program that uses a selection statement.

21. Write a C program that simulates a stock trading scenario for a novice trader. The program should take the initial investment amount, stock purchase price, and current stock value as inputs. It should then calculate and display the profit or loss based on the fluctuations in the stock's value.

22. Craft a program that takes the temperature in centigrade as input and outputs the corresponding weather message based on the given conditions as

Freezing weather, Very Cold weather, Cold weather, Normal temperature, Hot,Very hot

23. Write a program to take input for the employee's salary and gender, calculate the bonus based on the given criteria, and display the final salary that the employee will receive.

24. Your task is to create a program that prompts users to input their body temperature in Celsius. The program will then analyse the temperature provided and determine whether the individual has a fever based on the following criteria:
    1. If the body temperature is equal to or greater than 37°C, the program will indicate that the individual has a fever.
    2. If the body temperature is below 37°C, the program will indicate that the individual does not have a fever.
    3. Ensure the program handles valid input and provides appropriate feedback to the user.
    4. Display the result of the fever assessment to the user based on their body temperature input.

25. Craft a program that takes a positive decimal integer as input and transforms it into its hexadecimal and octal equivalents.

26. Implement a C program to determine the agent with the highest priority based on their success rate. The agent will be given the highest priority if they have a high success rate.

27. Craft a program that prompts the user to input 2 integer values. Calculate their product. If the product is even, divide it by 2 or divide it by 3 and display the result.

28. Develop a program that prompts users for a positive integer, checks if it's a perfect square using maths and conditionals, and outputs the result.

29. Craft a program that prompts the user to enter the age, annual tuition fee and total annual expenses.The program first asks for the user's age.If the age falls within the range of 18 to 25, Based on the comparison between the tuition fee and total expenses,

30. Craft a program to check whether a given character is an alphabet or not.

31. Craft a program that prompts users to input an integer.evaluate whether the input number is Even Number or Odd Number.

32. Develop a program that accepts a character input from the user. The program should then determine if the entered character is a vowel or a consonant or not a Letter.

33. Craft a program to determine whether a triangle is valid or not and if the triangle is valid then identify the type as Equilateral or Isosceles or Scalene.

34. Craft a program that prompts the user to input the number of hours worked and the hourly rate. It then calculates the total salary based on whether the hours worked are within the regular (40 hours) hours or exceed them. If the hours worked exceed the regular hours, it calculates the overtime pay for each hour exceeding the regular hours at 1.5 times the hourly rate. Finally, it displays the total salary.

35. Your task is to implement a program that calculates compound interest. And display the calculated compound interest to the user with precision 2.

36. Craft a program that takes a user-inputted integer and determines whether it is a buzz number or not. The program should use only conditional statements.

37. Craft a program that accepts the height of a person in centimeters as input and categorizes them based on their height into one of four categories: short ,average height, tall , very tall.

38. Craft a program that prompts the user to enter the number to be checked (num) and the divisor. checks if a given number is divisible by another number. Then, it verifies if the divisor is not zero. If it's not zero, it checks if the number is divisible by the divisor. Based on the result, it prints whether the number is divisible or not. If the divisor is zero, print an error message .

39. Craft a program that accepts input for a coordinate point and outputs the quadrant in which the point lies, or if it lies on the axis or at the origin.

40. Craft a program that prompts the user to input the speed limit and the speed of a vehicle. The program should then determine if the vehicle's speed exceeds the speed limit and issue a warning message if it does.

41. Craft a program that takes input from the user regarding the quantity of items purchased and the price per item. It then calculates the total cost of the purchase and discounts based on the total cost. Determine discounts based on the total cost.

42. Develop a program that prompts the user to input miles driven and gallons used, calculates MPG, and classifies gas mileage. The program should output the calculated MPG and its classification as Poor, Average, Good, or Excellent based on given ranges.

43. Craft a program that Takes input for the total score and total overs played by the batting team. Calculates the run rate per over and Converts the run rate to a percentage. Determines the likelihood of the batting team winning the match based on the percentage:
    - If the percentage is less than 50, it predicts that the opponent team can win.
    - If the percentage is exactly 50, it predicts that both teams have an equal chance of winning.
    - If the percentage is greater than 50, it predicts that the batting team can win.

Print the final output as percentage and prediction.

44. Create a program that assesses engine health based on inputted temperature (°C), oil pressure (PSI), and RPM. If temperature is above 75°C, oil pressure is below 12 PSI, or RPM exceeds 7000, engine health is poor. If temperature is above 50°C, oil pressure is below 12 PSI, or RPM exceeds 4000, engine health is fair.

45. Craft a program that calculates the discount amount based on the purchase amount. Apply different discount rates for different purchase ranges and Display the calculated discount amount.

46. Craft a program that prompts the user to input the coordinates (x, y, z) of the point.Then, it evaluates different conditions to determine the location of the point within the Cartesian coordinate system.
    - If all coordinates (x, y, z) are zero, print "origin"
    - If x and y coordinates are zero, print "z-axis"
    - If x and z coordinates are zero, print "y-axis"
    - If y and z coordinates are zero, print "x-axis"
    - If the x coordinate is zero, print " yz-plane"

47. Create a program to calculate the annual increment of an employee based on their performance rating. Assume that the performance rating is on a scale of 1 to 5, where 1 represents poor performance and 5 represents excellent performance.

48. Design a program that calculates the total cost of tickets for a movie theater. The ticket price depends on the age group of the person: Implement an additional discount of 11% if the number of tickets purchased is more than 8.

49. Craft a program that Prompt the user to enter the month (1-12) and select their choice.Calculate and display the approximate sunrise and sunset times based on the following criteria:
    - For months January, February, October, November, December:
    1) Sunrise: 7:00 AM
    2) Sunset: 5:30 PM
    - For March:

1) Sunrise: 6:30 AM
2) Sunset: 6:30 PM

- For months April, May, June, July, August:

1) Sunrise: 5:30 AM
2) Sunset: 7:30 PM

- For September:

1) Sunrise: 6:30 AM
2) Sunset: 6:00 PM

50. Write a program to find Get the lion's age as input. Also the maximum age of a lion is 26 years old. Display "Cub, Juvenile, Subadult, Young adult, Old adult with two precision .

51. Craft a program that takes input in the format "DD MM YYYY". Ensure that the input for day, month, and year are valid integers within reasonable ranges.Based on the input date, calculate the corresponding season.Divide the year into four quarters (Jan-Mar, Apr-Jun, Jul-Sep, Oct-Dec).Calculate the total number of days passed in the quarter up to the given date.
    Determine the season based on the total number of days passed:
    - Spring: March 1st to May 31st
    - Summer: June 1st to August 31st
    - Autumn: September 1st to November 30th
    - Winter: December 1st to February 28th (or 29th for leap years)

52. Craft a program that prompts the user to enter two characters.Calculate the ASCII values of these characters. Determine the relationship between the characters:
    - If both characters are vowels, calculate the sum of their ASCII values.
    - If one character is a vowel and the other is a consonant, calculate the difference of their ASCII values.
    - If both characters are consonants, calculate the product of their ASCII values.

53. Implement a BMI calculator that classifies individuals as Underweight, Normal weight, Overweight, or Obese (range within 0 to 100).

54. Write a program to get 3 inputs from the user. First get the selling price per unit. Next the quantity sold. Then The input Of middlemen was used. Choose from these middlemen A, B and C who deduct respectively. Find the total revenue after deduction by the middlemen. if the middlemen entered is A, B, C and print the original amount without deduction

55. Craft a program starts by prompting the user to input the angle of rotation in degrees and the direction of rotation (either clockwise or counterclockwise). Then, it asks for the initial x and y coordinates. Using trigonometric functions (cosine and sine), the program performs the rotation based on the specified angle and direction. Finally, it outputs the new coordinates after rotation.

56. Write a program to get two inputs from the user. If the two inputs are odd, then Multiply them and Display the output as a float with 4 decimal places.

57. Write a program to get a numeric input from the user. Check whether the given digit is a 5 digit number or not.

58. Craft a program that prompts the user to choose a shape (sphere, cube, or cylinder) and then calculates and displays the surface area and volume of the selected shape based on the provided dimensions.

59. Write a program to get two sides as input from the user. Check whether the inputs belong to square or rectangle and print the result as “Square” or “Rectangle” accordingly.

60. Craft a program should prompt the user to select a shape from the provided menu (cone, pyramid, or prism) by entering a corresponding number. After the user makes a selection, the program should request the necessary dimensions for the chosen shape, such as radius, base area, and height. Once the user provides the required dimensions, the program should calculate the

volume based on the selected shape and display the result with two decimal places.

61. Craft a program that takes a single character as input and determines whether it belongs to the set of vowels or consonants and finally decides if the input character is a vowel or a consonant or invalid.

62. Craft a program prompts the user to input the hour and minute values.It then calculates the angle between the hour (24 format) and minute hands based on the provided time.The angle is then adjusted to be within the range of 0 to 360 degrees.If the angle is negative, its absolute value is taken. Finally, the program displays the calculated angle.

63. Write a program that prompts the delivery vehicle operator to input the current coordinates (x1, y1) and the destination coordinates (x2, y2). The program then calculates and displays the Manhattan distance between the current location and the destination address. The Manhattan distance between two points (x1, y1) and (x2, y2) in a plane is calculated as the sum of the absolute differences of their Cartesian coordinates. Manhattan Distance = |x2 - x1| + |y2 - y1|. In other words, it's the distance a taxi would travel to get from one point to the other if it's only allowed to move horizontally or vertically.

64. Craft a program that prompts the user to input the base value and the exponent value, calculates the result of raising the base to the power of the exponent, and then prints the result.

65. Craft a program that prompts the user to input a frequency in Hertz. Read the frequency in integer and print the result as stated below with two decimal precision. Categorise the frequency as Infrasound, Audible sound and Ultrasound.

66. Your program should take the lengths of the two sides of the treehouse platform (base and height) and the length of the ladder as input. It should

then determine whether the lengths of the sides and the ladder form a right triangle according to the Pythagorean theorem(c^2 = a^2 + b^2).

67. The class is divided into groups, each consisting of two students. Each group is provided with a set of coordinates representing two points on the land survey. The students are tasked with calculating the slope between the two points using the provided coordinates.

68. Write a program to get age, gender and height. To get selected in a team a person should be either with height greater than or equal to 1.80 metre or should be a male with age being 18 or 18+

69. Develop a program to integrate into the school's educational software suite. The program should convert user-inputted digits to their word representations in lowercase (e.g., 0 as "zero", 1 as "one"). It should handle negative numbers by informing the user of their negativity.

70. Develop a program that requests input from users regarding a vehicle's size, fuel type, and intended purpose. The program should interpret inputs as follows: S for Small, M for Medium, L for Large in terms of size; G for Gasoline, D for Diesel, E for Electric for fuel type; and 3 for Personal Use, 1 for Commercial Use, 2 for Public Transport in terms of purpose. The program must validate inputs and provide corresponding information.

71. Craft a program that prompts the user to input a distance in either centimetres or metres. If the distance is greater than 1000, then the input is in metres and converts it from metres to kilometres. The program then outputs the converted distance accordingly.

72. Develop a seat booking program that categorises seats as available, reserved, or occupied based on a predefined range.

73. The students gather around their assigned workstations and input the x and y coordinates (x1, y1, x2, y2) into their Python program. The program

calculates the slope using the formula provided by the instructor: (y2 - y1) / (x2 - x1).

As the students input the coordinates and run the program, they observe the output based on the calculated slope:

- If the slope is positive, they understand that the land surface rises from the first point to the second point.
- If the slope is negative, they realise that the land surface descends from the first point to the second point.
- If the slope is zero, they conclude that the land surface is horizontal between the two points.
- If the denominator is 0, they recognize that the line between the points is vertical.

74. Write a program that takes two integer inputs from the user and swaps their values without using a temporary variable. Implement the swapping logic using arithmetic operations directly.
    Your program should:
    Prompt the user to enter two integer values.
    Read the input values.
    Swap the values of the two variables using arithmetic operations (addition and subtraction).
    Print the updated values of the variables after swapping.

75. Craft a program that prompts the user to input marks for 5 subjects, ensuring that the marks are within the range of 0-100. Marks must be between 0 and 100." Then, it checks if the student has passed in all subjects (scoring more than 35). If passed, it calculates the percentage of the student (displayed to two decimal places) and assigns a grade based on the percentage.

76. A program to determine the status of a battery (low, medium,Almost Full, full) based on its charge level in the range (0, 100). The charge level input

should be an integer. Handle cases where the input is outside the valid range (0-100).

77. Write a C program that calculates both kinetic and potential energy based on user input for mass, velocity, and height.
    Your program should:
    Prompt the user to choose between calculating kinetic or potential energy.
    1. Kinetic energy
    2. Potential energy

    If the user chooses to calculate kinetic energy:
    Prompt the user to enter the mass (in kilograms) and velocity (in meters per second).
    Calculate the kinetic energy using the formula: KE = (1/2) * mass * velocity^2.
    Display the calculated kinetic energy.
    If the user chooses to calculate potential energy:
    Prompt the user to enter the mass (in kilograms) and height (in metres).
    Calculate the potential energy using the formula: PE = mass * gravity * height, where gravity is approximately 9.8 m/s^2.

78. Create a program that determines an individual's eligibility to vote based on their age input, also categorising them into specific age groups such as Child, Teenager, Adult, Senior.

79. Design a program that prompts the user to input a value along with its unit, either in kilometres (k) or metres (m). The program should convert the entered value to the other unit and display the result with two decimal places.

80. Design a program that accepts an integer input from the user representing temperature, and then categorises and prints the weather condition based on predefined temperature ranges.

81. The textile owner orders the sales team to identify and classify the dress according to the size .The size of the dress listed below. So consider the list and help to classify it

82. Design a program that takes two integers, X and Y, from the user. Determine whether their sum is even or odd and print the result. Additionally, classify the integers as positive, negative, or one of each, and print the appropriate statement accordingly.

83. Write a program that prompts the user to input three integers, that finds and displays the largest and smallest numbers among them.

84. Write a program for checking sight condition of eye (power level is given as iput) based on the following conditions:
    1. The given value is 0 then normal.
    2. The given value is negative then print short sightedness
        a. The value is equal to or in between -1 to -4 then print minimum
        b. The value is equal to or in between -5 to -8 then print moderate
        c. The value is equal to or in between -9 to -12 then print high
        d. The value is equal to or in between -13 to -16 then print chronic
    3. The given value is positive then print long sightedness
        e. The value is equal to or in between 1 to 4 then print minimum
        f. The value is equal to or in between 5 to 8 then print moderate
        g. The value is equal to or in between 9 to 12 then print high
        h. The value is equal to or in between 13 to 16 then print chronic
    4. If the value given is more than the framed range then print invalid.

85. Craft a program that takes an integer 'n' as input and calculates the sum of even and odd digits separately. The integer 'n' is guaranteed to have four digits and to extract the rightmost digit and then shift the number to the right to access the next digit. Repeat this process for all four digits, updating the sums based on whether the digit is even or odd.

86. The task is to write a program that determines whether Janmansh will be able to complete all three assignments on time based on the time he starts working on them.

87. Design a program that prompts the user to input a seat number, determines if it's even or odd, and classifies its position in the theatre. If even, it checks if divisible by 4 and labels it as back of the theatre if true, otherwise as middle. If odd, it checks divisibility by 3 and labels it as back if true, otherwise as front.

88. Craft a program to calculate the total salary for Ninja, who has just received a job offer from a reputable company. The program should take inputs of Ninja's basic salary and grade, and then compute the total salary based on specified allowances and deductions including housing rent allowance (HRA), dearness allowance (DA), grade-based allowances, and provident fund (PF). Finally, the program should round off the total salary to the nearest integer and output the integral part.

89. Craft a program to determine the minimum number of changes needed to make five given integers equal. The program should consider scenarios where the integers are already equal, have only one distinct value, two distinct values, three distinct values, or four distinct values from 0 to 4.

90. The task is to write a C program that determines whether a turn is good based on the numbers rolled by Chef and Chefina on their dice.
Requirements:
    1. Prompt the user to input the numbers rolled by Chef and Chefina on their respective dice.
    2. Check if the sum of the numbers on the dice is greater than 6.
    3. Output "YES" if the turn is good and "NO" otherwise.
    4. Ensure proper input validation to handle invalid inputs, such as non-integer inputs or dice numbers outside the range of 1 to 6.

91. Develop a program to convert temperatures between Fahrenheit and Celsius scales.

92. write a C program to calculate the final amount a customer needs to pay after applying the discount based on the total amount spent on shopping.
Requirements:
    1. prompt the user to input the total amount spent by the customer (X).
    2. If X is less than or equal to 100, there is no discount.
    3. If $100 < X \leq 1000$, the discount is 25 rupees.
    4. If $1000 < X \leq 5000$, the discount is 100 rupees.
    5. If X is greater than 5000, the discount is 500 rupees.
    6. Ensure proper input validation to handle invalid inputs, such as non-integer inputs or total amounts outside the range of 1 to 10000.

93. Write a C program that prompts the user to input two characters. The program should display the addition of the ASCII value of the entered characters.

94. Design a program that calculates shipping costs for a shipping company based on package weight. The program should prompt the user to input the weight of the package and categorise it as small, medium, or large. Shipping costs are determined by small, medium, and large packages.

95. The task is to write a C program that takes a number as input and determines who takes it home among Alice, Bob, and Charlie based on their preferences.
(Note: Ensure proper input validation to handle invalid inputs, such as non-integer inputs or numbers outside the range of 1 to 1000.)
Requirements:
    1. For each test case, prompt the user to input a number (A).
    2. Check if the number meets Alice's preference criteria:
    3. If the number is even and a multiple of 7, Alice takes it home.
    4. Check if the number meets Bob's preference criteria:
    5. If the number is odd and a multiple of 9, Bob takes it home.
    6. If the number does not meet the criteria for Alice or Bob, Charlie takes it home.

7. Output the name of the person who takes the number home for each test case.

96. Alice's task is to write a program that takes the lengths of three sides of a triangle as input and determines whether the triangle is scalene or not. By completing these requirements, Alice will create a reliable program that accurately determines whether a given triangle is scalene or not.

97. Create a program that efficiently identifies the highest score among four component scores (two mid-term exam scores, a final exam score, and a project score) for each student in a school's grading system.

98. Design a program to facilitate ticketing for a theme park, automatically categorising ticket types based on age. The program should determine whether an individual qualifies for a child discount (age 12 or younger), and issue regular tickets for ages above 12.

99. Write a C program that presents a menu to the user to choose a shape (circle, rectangle, or triangle). The program should then ask for the necessary dimensions based on the user's choice and calculate the area of the chosen shape. The program should print the calculated area with two decimal precision.

100. You are tasked with developing a date validation and incrementation module for a project management software. The module should take user inputs for day, month, and year and validate whether the date is valid or not. If the date is valid, it should increment the date by one day and display the updated date.

101. Write a program that takes the coefficients of a quadratic equation ax2+bx+c =0 as input and calculates its roots.

Prompt the user to input the coefficients a,b and c of the quadratic equation.

1. If the discriminant is less than 0, print "The roots are imaginary"
2. If the discriminant is greater than or equal to 0, calculate the roots using the quadratic formula: r=(-b±d1)/2*a

102. Compute the total points earned using the formula: n + nn + nnn, where n is the transaction amount, nn is the transaction amount repeated twice, and nnn is the transaction amount repeated thrice. Print the total points earned.

103. Design a program for an ATM that enforces withdrawal limits based on account types. Regular accounts have a withdrawal limit of $300, while premium accounts have a limit of $500. If a user exceeds their account's limit, they should receive a corresponding message.

104. Design a program to calculate fines for overdue library books based on type and days overdue, charging $70/day for regular books and $150/day for reference books. If overdue by more than 77 days, add a $100 extra fine. Display the fine amount with two decimal precision and handle invalid inputs.

105. The program calculates a student's attendance percentage based on the total hours attended and total hours held in class. It then determines if the student is eligible to sit for the exam based on their attendance. If the percentage is 75% or higher, the student is allowed to sit for the exam; otherwise, they are not allowed. If the percentage is between 66% and 75%, they must apply for condonation.

106. Write a C program to calculate the charge for borrowing books from the library based on the number of days borrowed. Your program should prompt the user to input the number of days they intend to borrow books. Based on this input, calculate the total charge according to the provided fee structure and display the result.

107. Create a program that determines scholarship eligibility based on a student's GPA, extracurricular, and technical activities. If the student has a GPA of 3.5 or higher and participates in at least 3 extracurricular activities and 2 technical activities, they are eligible for the scholarship. Otherwise, they are not eligible.

108. Create a program to calculate the total rental cost based on the number of days, following specific pricing tiers: 2 units per day for 5 days or less, 3 units per day for 6 to 10 days, 4 units per day for 11 to 15 days, and 5 units per day for more than 15 days.

109. Create a program that prompts participants to enter their age and years of coding experience. Based on the given criteria, determine if they are eligible for the Junior or Intermediate division. Participants aged 13-18 with 1-3 years of experience qualify for Junior, while those with more than 3 years qualify for Intermediate.

110. You are tasked with creating a program that checks the validity of a triangle based on its side lengths. The program should prompt the user to input the lengths of the three sides of the triangle and then determine if the triangle is valid or not.

111. Create a program that determines the discount percentage and total bill amount for customers based on their total purchase amount and membership status. The program should apply different discount rates according to predefined criteria, considering whether the customer is a member or non-member.

112. Create a program that calculates transportation fare based on kilometres travelled. Prompt the user to input kilometres covered. Calculate fare according to specified conditions: $17 per km for <=10km, $17 per km for 13-100km, and $5 per km for >100km.

113. Integrate a C program into a digital clock system to convert 24-hour time to 12-hour format with AM/PM indication.

114. Write a program to swap two integers without using extra space or temporary variables. Use bitwise operators to achieve this.

115. Create a C program that determines the season based on user input of a month (1-12) and day (1-31). The program should output one of four seasons (Spring, Summer, Autumn, Winter) corresponding to the given date.

116. Develop a program to assign bonuses to players based on their chosen class. Warriors receive extra strength, Mages gain extra magic power, and Rogues are bestowed with extra agility.

117. Write a C program to calculate saving amount and percentage from total income after deducting other allowances. The program should take user inputs for total income and other allowances, then display the calculated saving amount and percentage.

118. Write a program that takes an integer input from the user and prints "Hey" if the input is divisible by both 8 and 1, "Hi" if divisible by 1, "Wow" if divisible by 8, and the input itself if not divisible by either.

119. Our task is to write a C program that prompts the user to enter a positive integer, calculates the sum of natural numbers up to the entered number using the sum_of_natural_numbers function, and prints the result. If the user enters a non-positive integer, the program should print an error message as “enter a positive integer”, and the condition should be less than or equal to zero.

120. You are developing a program to solve linear equations. Users will input coefficients 'a' and 'b' for the equation ax + b = 0. The program should handle various scenarios: if 'a' is zero and 'b' is zero, indicate infinite solutions; if 'a' is zero and 'b' is not zero, indicate no solution.

121. Craft a program that prompts the user to input a number. The program should then determine if the entered number is divisible by both 2 and 6.

122. Write a program that takes an identification number as input and determines whether it is a power of 2.

123. Write a C program that takes an integer input from the user representing the day of the week (1 for Monday, 2 for Tuesday, …, 7 for Sunday). The program should then print the corresponding day of the week.

124. Design a program to calculate the tax owed based on income levels, with varying tax rates depending on income brackets: 7% for incomes below $c, 15% for incomes between $a and $b valve, and 20% for incomes exceeding.

125. Craft a program that simulates a traffic light. Implement to take the current colour of the traffic light as input and print the corresponding message: "stop" for red, "proceed with caution" for yellow, and "Go" for Green.

126. Craft a program that prompts the user to input two numbers and then determines which one is greater, lesser, or if they are equal.

127. Craft a program to calculate and display the new salary of an employee based on their current salary and the percentage increment provided to them by the company.

128. Write a program that prompts library members to input the number of days they have exceeded the due date for their library books. The program should then calculate the total overdue charges based on the number of days and the fixed daily charge set by the library.

129. Design a program where students take turns calling out integers within a secret range. Depending on the integer called out, Ms. Li responds with 'Weird', 'Not Weird', or 'Invalid Input' following the specified conditions.

130. Create a program that takes a positive integer input from the user and analyses its last digit. Depending on whether the last digit is divisible by both 2 and 4, only by 2, or none of these, the program should print specific patterns.

131. Develop a program module for a construction company's salary management system to calculate and display daily wages based on the age and gender of employees.

132. Create a program that takes input for electric units consumed and calculates the bill based on given rates.

133. Given the total number of working days and the total number of days absent, calculate the percentage of class attended. If the attendance percentage is less than 75%, inquire if there is a medical cause (Y/N). If yes, display the attendance percentage.

134. Create a C program for a cola vending machine that sells Colas priced at $2 each. Users input the quantity desired and the amount of payment. The program checks if the quantity is valid and if there's sufficient stock. It then verifies the payment and dispenses the Colas accordingly,

135. Craft a program that reads a positive integer from the user and calculates the sum of its digits.

136. Craft a program that takes user input to enter the number of subjects and the scores for each subject. Calculate the total score by adding the scores of all subjects. Then, Calculate the CGPA by dividing the total score by the total possible score.

137. Write a C program that presents a menu to the user to choose a shape (circle, rectangle, or triangle). The program should then ask for the necessary dimensions based on the user's choice and calculate the area of the chosen shape. The program should print the calculated area.

138. The task is to write a C program that calculates the sum of the digits of a given number. The program should take an integer as input from the user, calculate the sum of its digits using a recursive function, and then display the result.

139. Accept the age, sex (‘M’ , ‘F’), number of days and display the wages accordingly

Age	Gender	Wage/day
>=18 and <30	M	700
	F	750
>=30 and <=40	M	800
	F	850

If age does not fall in any range then display the following message:
“Enter appropriate age”

140. Write a program that Accepts the market price from the user and calculate the net amount as (market price-discount) to pay according to the following criteria:

Market Price	Discount
>10000	20%
>7000 & <=10000	15%
>=2000 & <=7000	10%
<2000	No discount

141. Create a program that categorises temperatures as Very Cold (-20 to 0), Cold (1 to 10), Mild (11 to 20), Warm (21 to 30), Hot (31 to 40), Very Hot (41 to 50).

142. Implement a BMI calculator that classifies individuals as Underweight (<18.5), Normal weight (18.5-24.9), Overweight (25-29.9), or Obese (30 and above).

143. Write a program that takes a degree as input and outputs the corresponding cardinal direction (North, East, South, West) based on the degree range

144. Implement a weather type identification program that categorises weather conditions as Sunny, Cloudy, Rainy, or Stormy based on user input.

145. Develop a program that classifies triangles as Equilateral, Isosceles, or Scalene based on user input of three side lengths.

146. Create a program that categorises voltage levels as Low (0-50), Medium (51-100), High (101-150), or Critical (151 and above).

147. Implement a program that checks if a vehicle's speed exceeds the speed limit and issues a warning if the violation occurs.

148. Develop a program that calculates and classifies gas mileage as Poor (<15 mpg), Average (15-25 mpg), Good (26-35 mpg), or Excellent (36 and above).

149. Create a program that categorises electromagnetic waves into radio waves, microwaves, infrared, visible light, ultraviolet, X-rays, or gamma rays based on frequency.

150. Develop a program that diagnoses the health of a car engine based on temperature, oil pressure, and RPM within specified ranges.

151. Create a program that calculates the discount amount based on the purchase amount and applies different discount rates for different purchase ranges.

152. Write a C program to determine if a given number is a prime number within the range of 100 to 1000.

153. You are an HR manager at a prestigious educational institution, and your task is to write a Python program to assess the eligibility of faculty members for a promotion.The institution has specific criteria, with teaching experience being a crucial factor.
Hint:
Experience categorization
AP I – up to 4 ,
AP II up to 7,
AP III – up to 10 ,
ASP –up to 13,
Professor - up to 20,
Head - more than 20

154. Create a C program to decide whether a provided integer is a perfect square in the range of 500 to 1000.

155. Develop a program to verify if a given floating-point number is positive and greater than 10.5 but less than 20.3.

156. Write a C program to check if a user-entered character is an uppercase letter within the ASCII range of 65 to 90.

157. Implement a program to determine whether a user-provided integer is divisible by both 7 and 11 in the range of 200 to 500.

158. Develop a program to ascertain if a given integer is an odd number between 300 and 600.

159. Create a C program to check if a user-inputted decimal is within the range of 30.5 to 40.2.

160. Write a program to determine whether a provided integer is a multiple of 3 but not a multiple of 5 within the range of 100 to 300

161. Implement a C program to check if a given number is a perfect cube in the range of 1000 to 2000.

162. Develop a program to verify if a user-inputted character is a lowercase letter within the ASCII range of 97 to 122.

163. Create a program to decide whether a provided integer is even and greater than 150 but less than 300.

164. Write a program to check if a given floating-point number is negative and less than -5.7 within the range of -10.0 to 0.0.

165. Implement a program to ascertain if a user-entered integer is a power of 2 within the range of 512 to 1024.

166. Write a program that takes the initial velocity, time, and acceleration as input. Calculate the final velocity using the formula

final_velocity = initial_velocity + (acceleration * time). Print the final velocity as a floating-point number with two decimal places.

167. Create a program to determine whether a user-provided character is a vowel within the ASCII range of 65 to 122.

168. Write a program that takes three angles of a triangle as input and determines the type of triangle. Initially verify whether the given triangle is valid or invalid by calculating the sum of angles. If the sum of the angles is greater than 180, print "Invalid" otherwise determine the type of triangle (Acute, Right, Obtuse).

169. Craft a program that prompts the user to take the choices of both players as input, which is a rock, paper, scissors game. The program output should determine the winner using conditional statements and values mentioned above(i.e. if the value of the input is greater than 2 is invalid input).

170. Develop a program to ascertain if a provided integer is divisible by 4 and 6 but not by 8 within the range of 200 to 800.

171. Create a C program to check if a user-inputted decimal is an integer within the range of 25.0 to 50.0.

172. Write a program to check if a given number is a palindrome within the range of 1000 to 9999.

173. Create a program that determines whether a given year is a leap year or not, considering the range from 1800 to 2100.

174. Develop a program to check if a number is multiple of 3, 6 and 9 or not .

175. Write a program to print the Fibonacci sequence up to the 20th term within the range of 0 to 100.

176. Implement a program to check if a given number is a power of two within the range of 1 to 1024.

177. Create a program to check if a given number is an Armstrong number within the range of 100 to 999.

178. Write a program to calculate and print the factorial of a given number within the range of 1 to 10.

179. Develop a program to check weather the given year is a century or not. If century print the resultant year by adding decade to the input.

180. Implement a program to convert a given character into its equivalent ascii value.

181. Write a program to calculate and display the GCD of two numbers.

182. Develop a program to check if the given number is a plaindrome or not . Assume the input will be inside the range from 1 to 20.

183. Write a program to determine the type of triangle (equilateral, isosceles, or scalene) based on three given side lengths.

184. Implement a program to calculate the result of a given base raised to a power within the range of 2 to 5.

185. Craft a program prompts the user to enter the rainfall amount in centimetres. After obtaining the rainfall amount in centimetres, the program converts it to millimetres by multiplying the input by 10. This conversion is based on the fact that 1 centimetre equals 10 millimetres. The result is stored in the variable rainfall_mm.

186. Write a program to solve and display the roots of a quadratic equation within the range of -10 to 10.

187. Create a program to convert a decimal number to its Roman numeral representation within the range of 1 to 50.

188. Implement a program to convert temperatures from Fahrenheit to Celsius within the range of 0 to 100.

189. Develop a program to check if a given character is a valid hexadecimal digit (0-9, A-F, a-f) within the ASCII range of 48 to 122.

190. You are writing a program to determine if a person is eligible for a discount at a theme park. If the person is 12 years old or younger, they get a child discount; otherwise, they get a regular ticket. Write the if-else statement for this scenario.

191. You are developing a grading system for a school. If a student's score is above 90, they get an 'A'; if it's between 80 and 90, they get a 'B'; between

70 and 80, they get a 'C'; between 60 and 70, they get a 'D'; and below 60, they get an 'F'. Write an if-else statement to assign grades based on the score.

192. You are developing a game where players can choose a character class. If a player chooses "Warrior," they get extra strength; if they choose "Mage," they get extra magic power; if they choose "Rogue," they get extra agility. Write an if-else statement to assign the appropriate bonuses based on the player's choice.

193. Develop a program that determines if a given century year is a leap year or not based on the divisibility condition.

194. Ask the user to enter a number that corresponds to a scientific measurement.To classify the input according to the following criteria, use if-else statements:

1. A message showing a negative measurement will be displayed if the value is less than 0.
2. Classify the measurement as low positive if it falls between 0 and 10 (inclusive).
3. If the value is between 11 and 50 (inclusive), mark it as a moderate positive measurement.
4. The measurement should be classified as strong positive if the value is higher than 50.
5. A message stating a neutral measurement will be displayed if the value is precisely 0.

195. The program should include a well-designed menu, allowing users to interactively choose the option for circle area calculation. The user experience should be engaging, with clear prompts for input. circle (A)=πr2.

196. Craft a program that fulfills the requirements of your transportation company client. The program should prompt the user to input the start and end times of a trip, then calculate and display the duration of the trip in hours and minutes.

197. You're at a supermarket and are comparing prices of the same product across three different brands. You want to quickly determine which brand offers the best deal which saves on the monthly grocery bill based on the discount

198. Craft a program that prompts the user to enter two integers, the task is to swap their values if X is greater than Y . Then determine if the sum of X and Y is even or odd, and ascertain if both values are positive, negative, or one of each.

199. Craft a program that prompts the user to enter an input and that calculates the total distance a person travels in a week based on the distance between their home and office.The formula calculates the total distance traveled in a week for a round trip from home to the office and back to home. The distance from home to the office is represented by the variable X. Multiplying X by 2 accounts for the round trip (to and from the office).

200. Craft a program that takes Jerry's speed (X), Tom's speed (Y), and the initial distance between them as input, and calculates whether Tom will be able to catch Jerry. Consider the scenario where Jerry starts at a different position than Tom.The task is to determine whether Tom, with his speed Y metres per second, will be able to catch Jerry, who dashes at X meters per second. You'll need to calculate the time it takes for Tom to catch Jerry based on their speeds and the initial distance between them.

News

Tata Communications and NVIDIA Forge Partnership to Boost AI Infrastructure in India

India's AI ambitions get a boost as Tata Communications and NVIDIA partner to provide advanced computing infrastructure and platforms for AI development and deployment.

Rashi Grover
19 Apr 2024 15:41 IST

Latest Stories

Greener Networks for India: Telecom Embraces Efficiency

Himachal Pradesh Villages Get Mobile

IBM.
real time with IBM Cognos Analytics
Learn more →

IBM looks to capitalise on untapped cloud opportunity

Big Blue is seeking to bring the cloud operating model to approximately 80 per cent untapped mission-critical apps which have not moved to the cloud due to compliance, costs or latency issues

By Srikanth RP — On Jan 16, 2020

CLOUD | FEATURES | NEWS

SIGNUP FREE NEWSLETTER

First name

Last name

Your email

I'm not a robot

reCAPTCHA
Privacy - Terms

SIGN UP